

Ingedata: Using infrastructure as a powerful performance argument

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ABOUT INGEDATA

Google Cloud helped
Ingedata offer cost-effective
and quality processing services to
clients. The platform encourages rapid
of innovative services based on container
technology, analyzing artificial intelligence



About Ingedata

Founded in 2005, Ingedata offers outsourcing services, providing added value in the fields of machine learning (startups, data science, and artificial intelligence projects), complex data processing (mass retail, science), and technical design (industry, building, and construction).

Industries: Other

Location: Hong Kong

Google Cloud results

- Enabled the global deployment of a new IT infrastructure in less than two weeks
- Freed up 40% of IT technicians from maintenance constraints, and kept the Information System Management workforce stable despite activity doubling in a year
- Provides differentiated services to clients' order placement, real-time processing supervision,

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BigQuery (<https://cloud.google.com/bigquery/>)
(<https://cloud.google.com/speech/>), Cloud

Cloud CDN (<https://cloud.google.com/cdn/>),
Compute Engine

AI Platform (<https://cloud.google.com/ai-platform/>)

Cloud Speech to Text
(<https://cloud.google.com/speech/>)

Persistent Disk

Cloud Storage (<https://cloud.google.com/storage/>)
Operations

Compute Engine
(<https://cloud.google.com/compute/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Persistent Disk
(<https://cloud.google.com/persistent-disk/>)

Operations
(<https://cloud.google.com/products/operations>)

TensorFlow (<https://www.tensorflow.org/>)

**and
predictive
analytics**

Ingedata (<https://ingedata.net/>) is a smart outsourcing agency (BPO) dealing with complex tasks based on large volumes of data. The company combines a technical expertise in its **clients** (<https://ingedata.net/nos-clients/>)' sectors of activity with a transversal vision across multiple business sectors. This wealth of available experience allows Ingedata to conceive new approaches to designing and providing its services.

The company works with over a hundred clients, including GroupM, Dentsu-Aegis, Cdiscount, Virtual Expo Group, Argus de la presse, Figaro Classifieds, Infopro Digital, Heuritech, Earthcube, and Coop Academy, among others. Ingedata is a company with a strong international culture, with offices and production facilities in Europe, Madagascar, the Philippines, and Thailand.

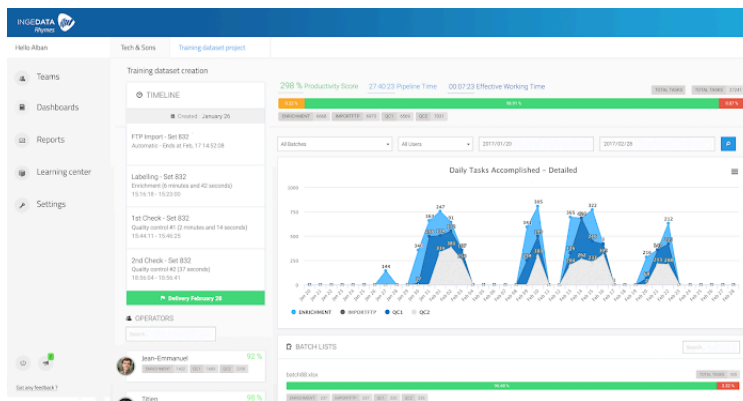
Ingedata offers its clients access to specialist teams totaling several hundred professionals across multiple areas including machine learning algorithms and their validation, the creation of datasets, product information management, and 3D modeling, among others. The company is able to quickly reconfigure its teams according to the needs of each project, making the cloud environment a perfect fit for Ingedata's company culture.

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—Yacine Petitprez, CTO,
Ingedata

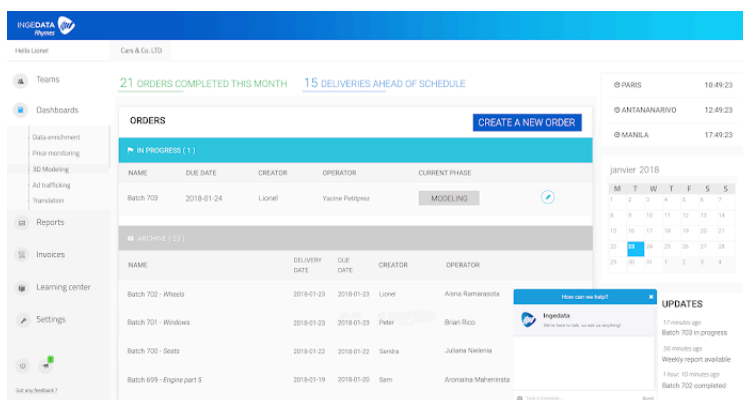
In 2016, Ingedata decided to move away from standard hosted services in order to migrate its entire IT system to Google Cloud (<https://cloud.google.com/>). Within two weeks, the company had transferred all of its data and applications to Compute Engine (<https://cloud.google.com/compute/>) and Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>)—virtual machines in the cloud based on Docker and Kubernetes technology. “It was all very simple. Very few scripts were required to complete the migration,” explains Yacine Petitprez, CTO at Ingedata.

“Since we implemented Google Cloud last year, Rhymes (<https://ingedata.net/notre-expertise/>) has functioned with 99.96% availability, which is much higher than what we were able to achieve with our previous provider,” says Yacine Petitprez. This move went hand in hand with the deployment of Rhymes, the new ERP (enterprise resource planning) solution developed by the company’s IT department. Rhymes is a hub for Ingedata’s activities including: processing of client tasks, client account management, human resources, and IT expansion—and it is available 24/7 across three continents.



Two steps ahead

“Initially, Rhymes was set up to meet a production need, but it ended up permeating the company’s culture to become a performance argument,” explains Titien Rico, Head of Sales and Marketing at Ingedata. On average, Ingedata uses 12 to 20 instances and 50 gigabytes of shared memory. Google Cloud automatically manages the balance of load, and the creation of pods (groups of Kubernetes containers) and nodes (clusters of virtual servers).



According to Yacine Petitprez, “Google’s container technology has enabled us to push Rhymes much further than we expected. Over and above the optimization of production, we can now offer our

clients the option of placing their orders directly via our information system, watching the progress of their processes in real time, accessing the dashboards, and benefiting from predictive analysis. Such transparency would have been impossible to achieve without Google Cloud."

In fact, we are no longer constrained by any technical limitations. We are absolutely certain that our information system will cope with new developments and projects, regardless of the volumes they involve. We are able to concentrate on what makes us unique: innovation in functionalities, quality, and analysis."

Factors behind the choice

During its benchmarking of three different cloud service providers, Ingedata identified numerous fundamental criteria.

From a functional and legal perspective, Ingedata needed to better ensure that the data processed complied with regulatory frameworks on a global level, offer transparent pricing that could adapt to variations in activity, simplify the management of IT budgets, and eliminate hidden costs.

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On the technical side, it needed to be able to deploy new functionalities within Rhymes daily without any break in service; to cope with any variation in the volume of processing; to activate new machines on demand for new project teams consisting of up to 70 specialists; and to require very little maintenance.

“The strength of the Google brand, its proven expertise in terms of compliance, and its leading-edge technologies meant that the choice was obvious,” says Titien Rico.

Innovation in services

Thanks to the possibilities offered by Google Cloud, Ingedata is now able to continuously expand its portfolio of services. For its banking and insurance

clients, the company analyzes very large volumes of unstructured data with the help of BigQuery

(<https://cloud.google.com/bigquery/>) and Tableau (<https://www.tableau.com/>), a tool that provides data visualizations stored in the cloud or available in real time.

By using the TensorFlow (<https://www.tensorflow.org/>) development framework, the IT department has been experimenting with the artificial intelligence and auto-learning services of Cloud Machine Learning Engine (<https://cloud.google.com/ml-engine/>) “Neural networks algorithms improve our predictions on the millions of data points that we process for our clients,” explains Yacine Petitprez.

In order to remain a front-runner in the services it offers, Ingedata has developed a specific offering dedicated to machine learning and artificial intelligence training. It uses APIs to turn conversations into texts (Speech-to-Text (/speech-to-text)) and to analyze unstructured texts (Cloud Natural Language (<https://cloud.google.com/natural-language/>)).

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Testing and deployment on the fly

“Thanks to Google Cloud, we have more or less replaced administrative and maintenance functions. Our IT teams previously spent too much time on low-added-value tasks. Now they are working on developing Rhymes further,” says the CTO. “Google Kubernetes Engine, based on Kubernetes, is a solution that is cheaper, easier to use, and more powerful than its competition. Completely integrated, it controls the pods and nodes, as well as the Cloud CDN (<https://cloud.google.com/cdn/>) network, Google Cloud's operations suite (</products/operations>) , or Compute Engine (<https://cloud.google.com/compute/>), and the artificial intelligence APIs. It's a real pleasure to use them as these elements are so seamlessly integrated.”

Thanks to containers, Ingedata has gained in both speed and quality. Code testing takes place in production-like situations and deployment is automated. 700 tasks are processed each hour by the Ingedata sector experts: inputting client data

sets, production, quality control, and export to output batches.

This workflow and any changes to the processing rules constantly query the Postgresql databases.

Google Kubernetes Engine servers rescale automatically depending on the workload. The data is processed very quickly thanks to SSD disks and is easily secured every hour using differential back-up disks, which share the blocks of files in order to minimize disk space usage.

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Compliance and budgetary control

With the help of Buckets

(https://cloud.google.com/storage/docs/json_api/v1/buckets)

and Google Persistent Disk

(<https://cloud.google.com/persistent-disk/>), Ingedata stores its clients’ assets by geographical zone in compliance with national regulations such as the

GDPR

(https://en.wikipedia.org/wiki/General_Data_Protection_Regulation)

or the Sarbanes-Oxley Act

(https://en.wikipedia.org/wiki/Sarbanes%E2%80%93Oxley_Act)

. These assets can take different forms: files, attachments, images, videos, reports, or elements taken from chatbots or sent by email.

Google Cloud Launcher

(<https://cloud.google.com/launcher/>) facilitates the connection between Google Cloud and the software used to support development, update blogs, and integrate devices. “Google Cloud eliminates all the hidden costs of infrastructure: the rates are

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”Placing our trust in an expert such as Google means efficiency, differentiation, and profitability.”